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Task Description:	We are studying nanometer sized aerosols that aggregate and thus become so dense that they eventually form a gel. Gelation of an aerosol is very novel and we discovered this phenomenon. We will study the kinetics of dense aerosol aggregation and gelation. we will do this with both experiments and computer simulations and compare these techniques. We will stuy the properties of the aerosol gels formed and develop their potential as useful materials. Aerosols are a very common "phase" of matter, from aeosols in our atmosphere that effect our global cimate to other planetary almospheres, such as the sand storms of Mars or the thick atmosphere of Saturn's moon Titan. All aerosols are		

inherently unstable and hence aggregate with time. Thus it is important to understand the mechanisms of aggregation and the structures that result. This understnding, which we are pursuing with our research, will help us deal with various planetary atmospheres as we explore the solar system. A second source of significance of our work is the unexpected discovery in our lab of novel ultralow density materials formed during aerosol gelation. These materials, which we call" aerosol gels", are similar to well-known aerogels but easier to make. Hence with further research may prove to have great technological utility.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

As described above, our research lays a fundamental understanding of aerosols which affect our global climate and visibility. Our discovery of aerosol gels may have great technological utility.

Superaggregates in Dense Particulate Systems
Flames. We have shown that soot formed in laminar diffusion flames of heavily sooting fuels evolves through four distinct growth stages which give rise to four distinct aggregate fractal morphologies [1]. Each successive stage grows upon the previous stage; hence aggregates of one morphology, parameterized by a fractal dimension, form larger superaggregates or gels of a different morphology with a different fractal dimension. These results were inferred from large and small angle static light scattering from the flames, microphotography of the flames, and analysis of soot sampled from the flames. These results and the analysis were substantiated by comparison to computer simulations. The growth stages occur approximately over four successive orders of magnitude in aggregate size and involve either diffusion limited cluster aggregation or percolation in either three or two dimensions.

Simulations. These results can be understood with the aid of our simulations, from the previous year, of off-lattice cluster-cluster aggregation for diffusion-limited (DLCA), ballistic-limited, and reaction-limited cluster aggregation classes [2]. We find that as the system evolves and becomes dense, the largest cluster develops a hybrid structure with mass fractal dimension Df =2.6 over large length scales, while at smaller length scales, the early time dilute-limit fractal structure with D=1.8 is frozen in. Consistent with our experiments the largest cluster is thus an aggregate of smaller aggregates with a different fractal dimension, and we call it a "superaggregate." The crossover length separating the two morphologies, which we call the critical radius of gyration, can be calculated based on a simple theory that assumes a monodisperse cluster size distribution. This agrees well with simulation results for DLCA.

Chamber Explosions. From digitized pictures of soot clusters formed after the explosion of a hydrocarbon gas mixed with oxygen, the cluster morphology was determined by two different methods: structure factor and perimeter analysis. Here again we found a hybrid, superaggregate morphology characterized by a fractal dimension of D = 1.8 between the monomer size, ca. 50 nm, and 1 micron and D = 2.6 at larger length scales up to ~ 10 micron. The superaggregate morphology is a consequence of late stage aggregation in a cluster dense regime near a gel point.

Aerosol Gels. We discovered a novel one-step method to produce aerogel-like porous materials [3,patent]. This method involves the gelation of nanoparticles (primary particles) in the aerosol phase to create a material that we have named an aerosol gel. So far we have made carbon aerosol gels via carbon nanoparticle gelation. However, any collection of finely divided primary particles with large enough volume fraction can produce an aerosol gel when allowed to aggregate regardless of the chemical composition of the parent primary particles. In the case of the carbon aerosol gel the initial aerosol is composed of nanometer sized carbon particles produced rapidly by exploding any one of a number of hydrocarbons with oxygen in a closed chamber. This method of making an aerosol gel is not a wet process and does not require a catalysis. Furthermore, the most important advantage of this method is the non-requirement of the supercritical drying step. The carbon aerosol gels have properties quite comparable to those of the carbon aerogels. The carbon aerosol gels have high specific area, an extremely low density, high electrical conductivity and highly crystalline primary particles. This carbon aerosol gel is significantly different than ordinary carbon black and soot formed as byproduct during combustion of any hydrocarbon fuel; it is a new material.

We have recently, since summer 2006, made aerosol gels of silica by exploding silane with any of a number of oxidizers. This proves our contention that any aerosol if dense enough will gel. We also expect our silica aerosol gels will rival silica aerogels in technological usefulness.

Aggregation Kinetics

Cluster-Cluster aggregation has been studied in our work via large scale Monte Carlo simulations for diffusion-limited (DLCA), ballistic-limited (BLCA), and reaction-limited (RLCA) cluster aggregation classes in three dimensions (3d) [4]. We investigated the growth kinetics for systems which evolve from a dilute to a dense state, a state we have name “cluster dense,” and then ultimately gel. We studied over three orders of magnitude in monomer volume fraction fv. DLCA model results show a power law growth kinetics with growth accelerating as the volume fraction occupied by clusters increases with time and the system becomes dense. We found that only a weak spatial correlation develops among clusters when dense, and cluster-cluster correlation is not sufficient to explain the late time enhancement of the growth kinetics. Instead, both exponents, z and lambda, characterizing the kinetics and size distribution, respectively, show universal functionality with cluster volume fraction, independent of the initial monomer volume fraction. Remarkably, the relationship between z and lambda maintains its mean-field nature in 3d. Results for the BLCA model in 3d is quite similar to the DLCA model while the 3d RLCA model shows an early time exponential growth that crosses over to a power law growth at late times when the system becomes dense.

The past year (2006) we have studied the aggregation of dense aerosols with light scattering. We built a small chamber 5.0 cm in diameter and 1.0 cm thick. In this chamber an acetylene/oxygen mixture was exploded to create aerosols with volume fractions of 10^-4 and greater. Both static and dynamic light scattering was performed. Static light scattering yielded kinetics that were faster than DLCA in the normal, what we call cluster dilute, regime. This was expected from our previous simulations for these cluster dense systems. We also observed an increase of the coagulation kernel homogeneity in accord with our simulations. Finally we saw a hint of a structural cross over indicative of superaggregates consistent with our simulations. These experimental verifications of our theory give a complete picture of aggregation in dense dispersions that ultimately form gels.

Task Progress:

Dynamic light scattering experiments are still under scrutiny, but indicate long nonexponential relaxations similar to those seen in liquid phase gels.

Electron microscope studies of soot aggregates from dense aerosols have shown the existance of superaggregates with a hybrid morphology as predicted by our simulations[7].

We have also studied an off-lattice Monte Carlo simulation of DNA-mediated colloidal assembly [5]. In this simulation, aggregation-fragmentation of a binary mixture of DNA-coated colloidal particles is studied through a simplified model of base-pair hybridization. Bonding between monomers is modeled as a simple temperature sensitive A/B type interaction, where type A and B monomers can only bond to the opposite type (no A/A or B/B attachments are allowed). The actual chemistry of base-pair hybridization is not included in the model. The morphological structures of the clusters formed as well as the kinetics of growth are analyzed in our two-dimensional simulations. The fractal dimension and kinetic growth exponents for clusters formed near the DNA 'melting' temperature agree with those seen previously for two-dimensional diffusion-limited cluster aggregation (DLCA) models. The clusters appear more compact, exhibiting signs of local order at intermediate temperature values. At higher temperatures, formation of large clusters is not favorable under the action of temperature-dependent fragmentation and the system eventually reaches a steady state as a collection of small aggregates. The temperature profile for this dissolution of the colloidal assembly is sharp, indicating that the selective hybridization process provides a highly sensitive measurement tool. At high temperatures, we analyze the steady state behavior of the average cluster size in terms of an aggregation-fragmentation model.

Light Scattering

Mie Scattering. We continued [6] our previous studies of patterns in Mie scattering. Our tactic to use as an alternative to the traditional scattering angle theta in describing light scattering from a uniform dielectric sphere; that alternative is the dimensionless parameter qR, where R is the radius of the sphere, and k is the wave number of the incident light. Simple patterns appear in the scattered intensity if qR is used in place of theta that have not been previously recognized. These patterns are characterized by the envelopes approximating the scattered intensity distributions and are quantified by the phase shift parameter where m is the real refractive index of the sphere. We found new patterns in these envelopes when the scattered intensity is normalized to the Rayleigh differential cross section. Mie scattering is similar to Rayleigh scattering when and follows simple patterns for which evolve predictably as a function of . The curves have a semi-quantitative universality with the phase shift parameter. These patterns allow us to present a unifying picture of the evolution of Mie scattering for changes in kR and m.

Multiple Scattering. We showed that the extent and effect of multiple scattering on angularly resolved light-scattering intensity measurements, the optical structure factor, can be quantitatively described by a single parameter, the average number of scattering events along the scattering volume [8]. This quantity is easily measured or calculated and hence provides a useful experimental indicator of multiple scattering, which is a hindrance to accurate structure factor measurements.

References

1. "Aggregates, Superaggregates and Gel-Like Networks in Laminar Diffusion Flames," W.G. Kim, C.M. Sorensen, D. Fry and Amit Chakrabarti, J. Aerosol Science 37, 386 (2006).

2. "Structural Crossover in Dense Irreversibly Aggregating Particulate Systems," D. Fry, A. Chakrabarti, W. Kim, C.M. Sorensen, Phys. Rev. E 69, 061401-10 (2004).

3. "Aerosol Gels: New Light Weight, High Surface Area Aerogel-Like Materials made via an Aerosol Process," R. Dhaubhadel, G. Gerving, A. Chakrabarti and C.M. Sorensen, Aerosol Sci. Tech. (submitted).

4. "Kinetics of Aggregation in Dense Particulate Systems, D. Fry, F. Pierce, C.M. Sorensen, and A. Chakrabarti, Phys. Rev. E (submitted).

5. "Aggregation-Fragmentation in a Model of DNA-Mediated Colloidal Assembly," F. Pierce, C.M. Sorensen, A. Chakrabarti, Langmuir 21, 8992 (2005).

6. "Patterns in Mie Scattering: Evolution when Normalized by the Rayleigh Cross Section," M.J. Berg, C.M. Sorensen, and A. Chakrabarti, Applied Optics 44, 7487-7493 (2005).

7. "Hybrid Superaggregate Morphology as a Result of Aggregation in a Cluster Dense Aerosol," R. Dhaubhadel, F. Pierce, A. Chakrabarti, and C.M. Sorensen, Phys. Rev. E 73, 011401 (2006).

8. "Multiple Scattering Effects on Optical Structure Factor Measurements", T Mokhtari, C. M. Sorensen, and A. Chakrabarti, Applied Optics 44, 7858 (2005).

Patent: Creation of Ultralow Density Porous Materials via Aerosol Gelation: Aerosol Gels, patent pending.

Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Peer-reviewed Journals	Pierce F, Sorensen CM, Chakrabarti A. "Aggregation-fragmentation in a model of DNA-mediated colloidal assembly." Langmuir. 2005 Sep 27;21(20):8992-9. PMID: 16171321 , Sep-2005
Articles in Peer-reviewed Journals	Kim WG, Sorensen CM, Fry D, Chakrabarti A. "Soot aggregates, superaggregates and gel-like networks in laminar diffusion flames." Journal of Aerosol Science. 2006 Mar;37(3):386-401. http://dx.doi.org/10.1016/j.jaerosci.2005.05.022 , Mar-2006
Articles in Peer-reviewed Journals	Dhaubhadel R, Gerving C, Chakrabarti A. Sorensen CM. "Aerosol gelation: synthesis of a novel, lightweight, high specific surface area material." Aerosol Science and Technology, 2007 Aug;41(8):804-10. http://dx.doi.org/10.1080/02786820701466291 , Aug-2007
Articles in Peer-reviewed Journals	Berg MJ, Sorensen CM, Chakrabarti A. "Patterns in Mie scattering: evolution when normalized by the Rayleigh cross section." Appl Opt. 2005 Dec 1;44(34):7487-93. PMID: 16353823 , Dec-2005

Articles in Peer-reviewed Journals	Dhaubhadel R, Pierce F, Chakrabarti A, Sorensen CM. "Hybrid superaggregate morphology as a result of aggregation in a cluster-dense aerosol." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Jan;73(1):011404. PMID: 16486137 http://dx.doi.org/10.1103/PhysRevE.73.011404 , Jan-2006
Articles in Peer-reviewed Journals	Mokhtari T, Sorensen CM, Chakrabarti A. "Multiple scattering effects on optical structure factor measurements." Appl Opt. 2005 Dec 20;44(36):7858-61. PMID: 16381538 , Dec-2005